# Bridging Text and Graph Topology for Wikipedia Article Recommendation

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**Abstract.** Article recommendation systems are commonly evaluated on benchmark graph datasets such as Cora, CiteSeer, PubMed, and Wiki-CS. Among these, Wiki-CS is widely used due to its open accessibility and graph-ready structure. However, its associated article content is outdated and does not adequately reflect recent developments in computer science, limiting its usefulness for modern text-aware recommendation methods. To address this limitation, we reconstruct and update Wiki-CS by aligning article titles, recovering article texts through the integration of the Wikipedia STEM corpus and the Wikipedia API, and generating semantic representations using state-of-the-art embedding models. The resulting benchmark provides complete textual coverage for all active nodes and supports reproducible recommendation experiments with contemporary Transformer-based language models. Using this reconstructed dataset, we develop an article-to-article recommendation framework that combines semantic and structural information. In particular, we investigate degree-bias-corrected Node2Vec variants that reduce the influence of highly connected nodes in graph embeddings, alongside hybrid retrieval approaches that integrate textual and graph-based representations with a pooling operation. We benchmark semantic, structural, hybrid, and GNN-based methods under a revised evaluation protocol. Experimental results show that hybrid retrieval models consistently outperform single-source baselines, providing a more robust balance between semantic relevance and structural connectivity.

**Keywords:** Article recommendation · Graph recommendation · Hybrid retrieval · Wikipedia graphs · Benchmark reconstruction

## 1 Introduction

Wikipedia is a globally used, openly accessible knowledge environment whose articles can be read both as natural-language documents and as vertices in an explicit hyperlink network. Its pages are publicly reachable as text, its links encode relations among topics, and its coverage spans a broad range of domains. For these reasons, Wikipedia has repeatedly been used in benchmark construction, retrieval studies, graph learning, knowledge-intensive NLP, and content-oriented

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analysis [12, 13, 22, 14, 23]. From an analytical perspective, this dual availability of content and relational structure makes the same article collection suitable for semantic modeling, structural modeling, and experiments that combine both views.

Within such an environment, article-to-article recommendation is a natural and practically useful problem. Given a page currently being examined, the task is to recommend other pages worth reading next, ideally preserving topical relevance while still broadening navigation. Wikipedia's hyperlink graph offers a strong starting point because direct links provide an explicit structural signal between articles [13, 5]. At the same time, relying only on graph structure may be too restrictive. For example, a graph-driven recommender around an operating-systems article may prioritize directly linked hub pages such as Linus Torvalds, while semantically close but less directly connected pages on kernel design, Unix-like systems, or file-system concepts remain under-ranked. This motivates article content as a second source of evidence rather than a mere auxiliary feature.

This dual structure also makes Wikipedia scientifically useful. Openly available graph extractions make structural experimentation feasible, and resources such as WikiLinkGraphs and Wiki-CS support large-scale studies of Wikipedia connectivity [3, 12]. However, the open resources are uneven across structure and content. Wiki-CS was introduced for node classification rather than article recommendation, and it does not directly expose the recovered, up-to-date raw article text needed to regenerate semantic representations with modern encoders. Graph-text paired datasets such as WikiGraphs are valuable for related objectives [22], yet they do not provide the same Wiki-CS article set with complete text for recommendation. As a result, there remains a gap between the availability of open Wikipedia graphs and the availability of an open benchmark that supports joint structural and semantic recommendation experiments on the same articles.

The first goal of this study is to close that gap at the dataset level. We focus on Wiki-CS because its computer-science article subset provides a topic-bounded setting in which recommendation consistency can be examined within a coherent domain. Starting from this benchmark, we reconstruct the article layer by cleaning titles, recovering raw text from open sources, and attaching modern dense embeddings, thereby producing an updated benchmark that supports contemporary semantic retrieval.

The second goal is methodological. We study recommendation models that use graph topology, article content, or both, and we treat these signals as complementary rather than interchangeable. In particular, we investigate a hybrid recommendation framework that combines BERT-based semantic embeddings, instantiated with BGE-M3 [2], and Node2Vec-based structural representations [5], alongside semantic-only, structural-only, GCN, and GraphSAGE baselines [7, 6]. We also examine corrected structural variants and pooled semantic representations in order to better balance structural and textual evidence. To assess the proposed approach in a broader experimental setting, we compare it against a representative set of semantic, structural, hybrid, and GNN-based alternatives.

The contributions of this paper are threefold:

- We introduce Custom-WikiCS, a reconstructed and text-complete version of Wiki-CS with recovered raw text for every active node and support for modern embedding pipelines.
- We propose a hybrid article recommendation framework that combines BGE-M3 semantic representations with Node2Vec-based structural signals, including corrected structural scoring and pooled fusion variants.
- We provide a comparative evaluation across recommendation-accuracy and recommendation-quality metrics under a revised undirected protocol, covering semantic, structural, hybrid, and GNN-based baselines.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the methodology, including the Custom-WikiCS reconstruction and the recommendation framework. Section 4 describes the experimental setting and results. Section 5 concludes the paper.

## 2 Related Work

Wikipedia can be modeled naturally as a graph whose nodes are articles and whose edges are hyperlinks. This makes it a practical environment for graph representation learning and graph-based recommendation. Wiki-CS provides a well-established Wikipedia-derived graph benchmark for graph learning research [12], although its original use case is node classification rather than article recommendation. More directly related work includes WikiRecNet, which studies editor-oriented Wikipedia article recommendation using graph structure and representation learning [13]. Its personalized editor-to-article task differs from our article-to-article setting, but it demonstrates the value of structural and textual article representations at Wikipedia scale. One lesson from this line of work is that structural proximity is informative, but recommendation quality should not be reduced to a generic graph prediction task.

Textual embeddings provide a second major signal family. Transformer-based encoders [19] and BERT-style models [4] have made dense semantic retrieval practical at sentence and document level. Sentence-BERT [15] further showed how BERT-like architectures can be adapted for efficient similarity search. More recent retrieval-oriented models, such as BGE-M3 [2], are explicitly designed for strong dense retrieval across multilingual settings and thus fit the recommendation scenario considered here.

Structural embeddings capture graph position, local neighborhood, and community patterns directly from the edge structure. Node2Vec [5] is one of the most influential methods in this family, learning node representations from random-walk contexts. In recommendation settings, structural embeddings can recover relations that text alone may miss, although they can also become sensitive to hub effects and graph-specific artifacts if the evaluation protocol is not carefully aligned with the structural task.

Graph neural networks provide another structural modeling family. GCN remains a standard transductive baseline [7], while GraphSAGE [6] introduces neighborhood aggregation with an inductive flavor. Survey work on GNNs for text-related settings suggests that these models remain central at the interface of graph structure and linguistic representation [21]. In parallel, multimodal recommender systems increasingly combine heterogeneous content signals rather than relying on a single modality [9].

The present work is positioned at the intersection of these directions. It uses Wikipedia as a graph recommendation environment, updates a widely used benchmark to support modern text modeling, and compares semantic, structural, hybrid, and GNN-based article recommendation models within one experimental framework.

## 3 Method

### 3.1 Custom-WikiCS Reconstruction

The original Wiki-CS graph is useful as a structural benchmark, but it is less convenient for modern article recommendation because the benchmark is graph-ready without being text-complete for current retrieval pipelines. In particular, the node set is easy to load as a graph, yet the benchmark does not directly expose the reconstructed raw article text needed to regenerate semantic features with modern encoders. For a recommendation setting centered on article-to-article retrieval, this is a practical limitation: text quality and text availability are not auxiliary metadata, but part of the representation being evaluated.

Custom-WikiCS addresses this limitation through a reconstruction pipeline with three main stages. First, article titles are cleaned and aligned in order to improve consistency across graph records, the Wikipedia STEM corpus[1], and API lookups. This stage resolves naming mismatches, formatting inconsistencies, and stale title variants that would otherwise prevent article recovery. Second, raw article text is recovered by combining the Wikipedia STEM corpus with Wikipedia API backfilling[2], producing a text-complete collection for the active article set. Third, the reconstructed collection is used to generate modern text representations, enabling semantic retrieval models and hybrid recommendation variants on top of the same graph. In the final reconstructed benchmark, every active node is associated with usable article text and can therefore participate in both structural and semantic evaluation.

After reconstruction and cleaning, isolated nodes are removed and the final graph contains the statistics reported in Table 1. The resulting benchmark supports both graph-structural experiments and modern text-based retrieval on the same article set.

[1] `https://www.kaggle.com/datasets/conjuring92/wiki-stem-corpus`, accessed 22 June 2026

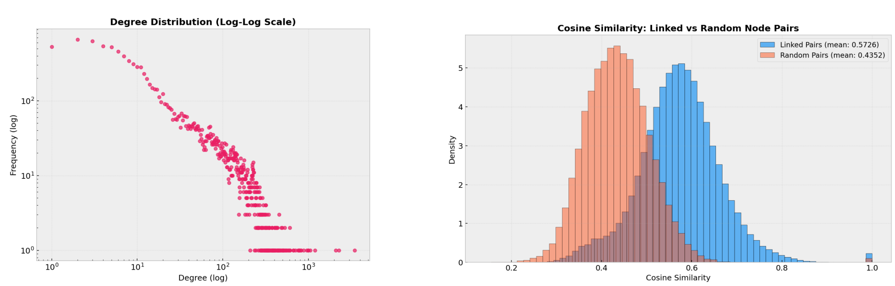
[2] MediaWiki Action API documentation: `https://www.mediawiki.org/wiki/API:Main_page`, accessed 22 June 2026

**Table 1.** Custom-WikiCS graph statistics after reconstruction and cleaning

| **Statistic** | **Value** |
|---|---|
| Initial nodes | 11,701 |
| Removed isolates | 334 |
| Active nodes | 11,367 |
| Directed edges | 291,039 |
| Undirected edges | 216,123 |

The cleaned graph exhibits a heavy-tailed degree distribution, indicating that a small number of hub articles coexist with a much larger set of lower-degree nodes. This matters for article recommendation because purely structural methods can become dominated by hubs if degree effects are left untreated. The degree statistics are also strongly skewed in the cleaned graph: total degree ranges from 1 to 3,474, with mean 51.21 and median 16. This gap between mean and median is consistent with a hub-dominated article network rather than a uniformly mixed graph. The left panel of Fig. 1 visualizes the same behavior on log-log axes and supports the power-law-style correction heuristic introduced later in the recommendation framework.

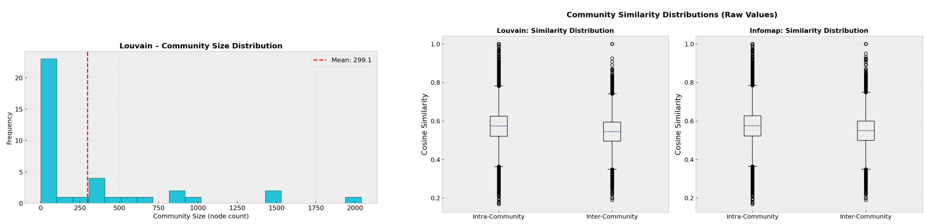
At the same time, the graph is semantically assortative. Linked article pairs are substantially closer in the semantic embedding space than randomly sampled pairs: under cosine similarity, the linked mean is 0.5726 while the random mean is 0.4352. This is important because it shows that hyperlink connectivity and textual meaning are aligned rather than independent. In other words, semantic retrieval is not being imposed on an unrelated graph; it operates on a benchmark where the link structure already carries topic-level coherence. The right panel of Fig. 1 summarizes this effect.




(a) Log-log degree distribution (b) Linked vs. random cosine similarity

**Fig. 1.** Structural and semantic diagnostics of Custom-WikiCS. The degree profile is heavy-tailed, while linked article pairs are semantically closer than random pairs in the BGE-M3 embedding space

Community analysis further shows that the graph is neither flat nor uniformly mixed. Louvain detection [1] yields modularity 0.6476 over 38 communities, with a largest community of 2,040 nodes and mean community size 299.13. This confirms that the cleaned graph contains broad topic regions rather than a single homogeneous article pool. Moreover, the semantic coherence of these communities is measurable: intra-community cosine similarity is higher than inter-community similarity under both Louvain (0.5729 vs. 0.5435) and Infomap [16] (0.5748 vs. 0.5480). This means that graph communities are not only topological partitions but also semantically coherent article groupings. Such a structure is useful for recommendation because it creates meaningful local neighborhoods beyond direct edge presence. Figure 2 summarizes the community-size distribution together with the intra/inter community similarity comparison.




(a) Louvain community size distribution

(b) Intra- vs. inter-community similarity

**Fig. 2.** Community diagnostics for Custom-WikiCS. The cleaned graph contains non-trivial communities, and those communities remain semantically coherent

### 3.2 Recommendation Framework

Let $G = (V, E)$ denote the reconstructed article graph. For a source article $u \in V$, the goal is to return a ranked list of candidate articles $v \in V \setminus \{u\}$ that are relevant to $u$ under semantic, structural, or hybrid similarity. The task is query-based and article-centric: the source article serves as the query, and the recommendation model ranks all remaining articles accordingly. The framework that we built for recommendation is figured in Fig 3. In the next part, we introduce the details of semantic and structural embedding generations and the proposed combination strategies for final recommendation.

**Semantic Retrieval** The semantic component uses BGE-M3 text embeddings [2]. Each article is represented by a 1024-dimensional dense vector derived from its reconstructed raw text. In the implementation used for this paper, embeddings are generated in dense mode only, cached for reuse, and encoded with a maximum input length of 2048 tokens per article. Long articles are therefore truncated during embedding generation, but every active node remains associated with a usable semantic representation. Recommendation scores in the

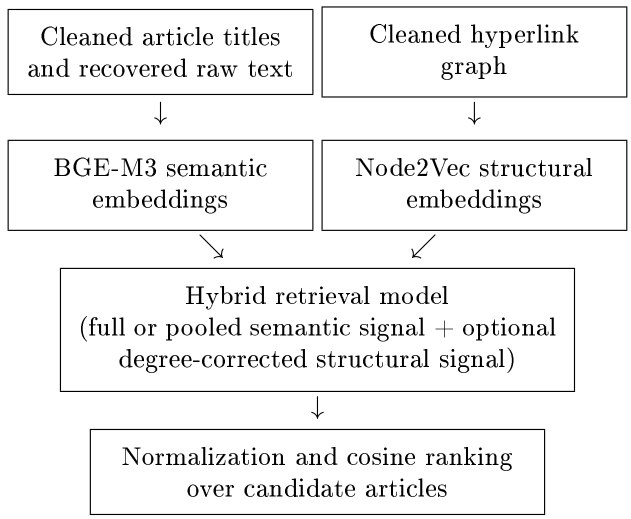



**Fig. 3.** Flagship hybrid recommendation pipeline used in the paper

semantic-only setting are computed by cosine similarity between source and candidate embeddings:

$$s_{\text{sem}}(u, v) = \cos(\mathbf{b}_u, \mathbf{b}_v). \tag{1}$$

This component captures semantic proximity even when the hyperlink relation is sparse or absent, and it serves as both a standalone baseline and one component of the hybrid models.

**Structural Retrieval** The structural component uses Node2Vec-style representations [5] computed on the Wikipedia hyperlink graph. In our implementation, these representations are obtained from unbiased random walks with $p = q = 1.0$, 10 walks per node, walk length 80, context window 5, and a final 128-dimensional embedding derived through an SVD-based co-occurrence factorization of the walk contexts. Two structural variants are considered. The first is an uncorrected Node2Vec configuration, which ranks candidates using cosine similarity in the structural embedding space:

$$s_{\text{str}}(u, v) = \cos(\mathbf{n}_u, \mathbf{n}_v). \tag{2}$$

The second is a corrected variant that applies a power-law degree penalty during structural scoring in order to reduce the tendency of high-degree hubs to dominate retrieval. We propose this correction for reducing a high-degree bias at the recommendation.

The correction factor for a node $v$ with undirected degree $d_v$ is defined as

$$p_v = \log(d_v + 1)^{\alpha}, \tag{3}$$

where the exponent is estimated from the structural training graph and is approximately $\alpha \approx 2.9966$ in the revised evaluation setting. This choice should be understood as a heuristic motivated by the heavy-tailed degree profile observed in the cleaned graph, where a power-law-like exponent near 3 provides a practical scale for suppressing hub dominance without removing structural similarity altogether. The logarithm softens the extreme upper tail of the degree distribution, while the power exponent preserves a stronger penalty for structurally over-connected nodes. The corrected structural score becomes

$$s_{\mathrm{corr}}(u, v) = \frac{\cos(\mathbf{n}_u, \mathbf{n}_v)}{p_u p_v}. \tag{4}$$

**Hybrid and GNN-Based Models** The central model family of this paper combines semantic and structural representations. We evaluate four main hybrid variants:

- direct concatenation of BGE-M3 and uncorrected Node2Vec;
- direct concatenation of BGE-M3 and corrected Node2Vec;
- pooled BGE-M3 with uncorrected Node2Vec;
- pooled BGE-M3 with corrected Node2Vec.

In the pooled variants, the 1024-dimensional semantic representation is reduced to 128 dimensions by block-wise pooling before concatenation, making the semantic and structural components more comparable in scale. This produces a 256-dimensional hybrid representation rather than the 1152-dimensional vector obtained by direct concatenation.

More explicitly, the non-pooled hybrid representation is formed as

$$\mathbf{h}_u = [\mathbf{b}_u; \mathbf{n}_u], \tag{5}$$

while the pooled representation uses a reduced semantic vector $\bar{\mathbf{b}}_u \in \mathbb{R}^{128}$ obtained from contiguous blocks of the original BGE-M3 embedding:

$$\mathbf{h}_u^{\mathrm{pool}} = [\bar{\mathbf{b}}_u; \mathbf{n}_u]. \tag{6}$$

In corrected hybrids, the same degree-normalized structural signal is used before concatenation. After construction, the hybrid vectors are normalized and ranked by cosine similarity against all candidate articles. The purpose of pooling is not to create a new semantic encoder or to claim semantic meaning for contiguous embedding blocks. It is used as a pragmatic dimensional balancing step so that the 1024-dimensional text representation does not numerically overwhelm the 128-dimensional structural contribution.

The benchmark also includes GCN and GraphSAGE structural baselines, as well as a ranking-level fusion baseline combining GCN and BGE-M3. Together, these models span semantic-only, structural-only, hybrid-concatenative, rank-fusion, and GNN-based retrieval families. The GCN and GraphSAGE baselines are trained as two-layer link-prediction models with negative sampling under the same revised protocol, using dropout 0.5, Adam optimization with learning rate 0.01 and weight decay $5 \times 10^{-4}$, and early stopping up to 200 epochs.

## 4 Experiments and Results

All main results reported in this paper use the revised undirected structural evaluation protocol. Starting from the undirected version of the cleaned graph, we construct a split with 216,123 undirected edges, of which 172,900 are used for training and 43,223 for testing. To align ranking evaluation with article-to-article retrieval, each undirected test edge $\{u, v\}$ is expanded into two directed queries, $u \to v$ and $v \to u$, yielding 86,446 evaluation pairs over 9,022 evaluation nodes.

This revised protocol plays an important methodological role. Alternative evaluations based on held-out directed edges are possible, but the comparison in this paper is intentionally centered on the undirected structural graph because Node2Vec neighborhoods, community analysis, and the degree correction heuristic are all defined over that structural view. The revised protocol should therefore be interpreted as a structurally aligned comparison setting for semantic, structural, and hybrid retrieval families, rather than as the only valid formulation of article recommendation. Directed navigation behavior remains meaningful and is left open for future extension.

The experimental workflow is notebook-centered and relies on cached structural and semantic artifacts generated from the reconstructed dataset. The benchmark includes semantic, structural, hybrid, and GNN-based models evaluated under the same underlying revised split, making the reported comparisons directly comparable at the level of ranking outputs. For ROC-AUC and PR-AUC, positive undirected test edges are paired with an equally sized sampled negative edge set so that the discrimination-oriented extension is measured on a matched binary comparison problem.

### 4.1 Evaluation Metrics

Let $Q$ denote the set of evaluation queries, let $r_q$ be the rank of the held-out target article for query $q$, and let $L_q$ be the returned recommendation list. Our ranking metrics follow standard information retrieval practice [10, 20]. Because each query has exactly one held-out target in the ranking protocol, several metrics simplify. The basic top-list metrics are

$$\text{Hit@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[r_q \leq K] \tag{7}$$

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{r_q} \tag{8}$$

For the same single-target setting, normalized discounted cumulative gain reduces to

$$\text{NDCG@}K = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathbf{1}[r_q \leq K]}{\log_2(r_q + 1)} \tag{9}$$

since the ideal DCG of one relevant item equals 1. Likewise,

$$\text{Recall@}K = \text{Hit@}K \tag{10}$$

and average precision becomes

$$\text{AP@}K(q) = \begin{cases} 1/r_q, & r_q \leq K, \\ 0, & r_q > K, \end{cases} \tag{11}$$

$$\text{MAP@}K = \frac{1}{|Q|} \sum_{q \in Q} \text{AP@}K(q) \tag{12}$$

In the main results table, the unlabeled NDCG, Recall, and MAP columns correspond to $NDCG$@100, *Recall*@100, and $MAP$@100. Therefore Recall is numerically identical to Hit@100 in our setup.

For the discrimination-oriented extension, ROC-AUC and PR-AUC are computed from model scores assigned to positive undirected test edges and sampled negative edges. ROC-AUC measures the area traced by true-positive rate and false-positive rate across thresholds, while PR-AUC measures the area under the precision-recall curve and is especially informative under class imbalance [17]:

$$\begin{aligned} \text{TPR}(\tau) &= \frac{TP(\tau)}{TP(\tau) + FN(\tau)}, \\ \text{FPR}(\tau) &= \frac{FP(\tau)}{FP(\tau) + TN(\tau)}, \\ \text{Precision}(\tau) &= \frac{TP(\tau)}{TP(\tau) + FP(\tau)}. \end{aligned} \tag{13}$$

We also report recommendation-quality metrics that characterize broader system behavior beyond target ranking. Let $C$ denote the candidate article set, let $d(v)$ be the undirected degree of article $v$, and let $M = \sum_{v \in C}(d(v) + 1)$. The Degree Popularity Ratio is computed as

$$\text{Degree Popularity Ratio} = \frac{\frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|L_q|} \sum_{v \in L_q} d(v)}{\frac{1}{|C|} \sum_{v \in C} d(v)} \tag{14}$$

This ratio is related to popularity lift in recommendation-bias evaluation [8], but it uses the candidate-pool mean as its baseline and deliberately retains a neutral value of 1, rather than the zero-centered definition of conventional lift. Values above 1 indicate that a model tends to recommend higher-degree articles than the candidate-pool baseline.

Aggregate Diversity counts how many distinct articles appear at least once across all recommendation lists [11, 24]:

$$\text{Aggregate Diversity} = \left| \bigcup_{q \in Q} L_q \right| \tag{15}$$

Novelty is measured with a graph-degree-based self-information score,

$$\text{Novelty} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|L_q|} \sum_{v \in L_q} -\log_2 \left( \frac{d(v)+1}{M} \right) \tag{16}$$

which is the graph-adapted form used in our implementation rather than an interaction-frequency definition [18]. For Intra-List Diversity (ILD), we average pairwise cosine distances within each recommendation list in semantic embedding space:

$$\text{ILD} = \frac{1}{|Q|} \sum_{q \in Q} \frac{2}{|L_q|(|L_q|-1)} \sum_{i<j,\ v_i, v_j \in L_q} (1 - \cos(e(v_i), e(v_j))) \tag{17}$$

where $e(v)$ denotes the BGE-M3 embedding of article $v$ and the formula is applied to lists with at least two items [24]. Finally, let $c_{(1)} \leq \cdots \leq c_{(n)}$ denote the sorted recommendation counts of the distinct articles that appear in at least one list. The Exposed-Item Gini is

$$\text{Exposed-Item Gini} = \frac{\sum_{i=1}^{n} (2i - n - 1) c_{(i)}}{n \sum_{i=1}^{n} c_{(i)}} \tag{18}$$

which summarizes how unevenly recommendation exposure is distributed among items that receive at least one recommendation [24]. Because zero-exposure candidates are excluded, this is not a full-catalog Gini; Aggregate Diversity separately reports how much of the catalog is reached. Higher Hit-based metrics, MRR, NDCG, MAP, ROC-AUC, and PR-AUC indicate better target recovery or discrimination, while higher Aggregate Diversity, Novelty, and ILD indicate broader coverage, less-popular recommendations, and greater within-list semantic diversity, respectively. A lower Degree Popularity Ratio indicates less hub-oriented recommendations, and a lower Exposed-Item Gini indicates more even exposure among reached items.

### 4.2 Results

**Combined Metrics on the Revised Protocol** Table 2 summarizes the main revised evaluation results. The revised protocol highlights clear strengths for the corrected hybrid family in top-list recommendation metrics, while also revealing metric-dependent trade-offs across semantic, structural, and hybrid families.

**Table 2.** Combined ranking and recommendation-quality metrics under the revised evaluation protocol

| Model | H@1 | H@10 | H@50 | H@100 | MRR | NDCG | Recall | MAP | Deg. Pop. Ratio | Exp. Gini | Agg. Div. | Novelty | ILD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGE pooled + Node2Vec (corrected) | 1.03% | 8.67% | 31.46% | 47.81% | 0.0410 | 0.1160 | 0.4781 | 0.0392 | 1.3301 | 0.4110 | 11,326 | 14.2738 | 0.4372 |
| BGE-M3 + Node2Vec (corrected) | 1.07% | 8.67% | 32.41% | 49.81% | 0.0421 | 0.1202 | 0.4981 | 0.0403 | 1.3081 | 0.4127 | 11,268 | 14.2783 | 0.4316 |
| BGE pooled + Node2Vec (uncorrected) | 0.76% | 7.83% | 32.64% | 50.74% | 0.0376 | 0.1176 | 0.5074 | 0.0358 | 1.3397 | 0.4613 | 11,186 | 14.0351 | 0.4558 |
| BGE-M3 + Node2Vec (uncorrected) | 0.76% | 7.83% | 32.64% | 50.74% | 0.0376 | 0.1176 | 0.5074 | 0.0358 | 1.3400 | 0.4605 | 11,164 | 14.0340 | 0.4557 |
| Node2Vec (uncorrected) | 0.75% | 7.81% | 32.52% | 50.60% | 0.0375 | 0.1172 | 0.5060 | 0.0357 | 1.3374 | 0.4482 | 10,966 | 14.0466 | 0.4569 |
| Node2Vec (corrected) | 0.75% | 7.81% | 32.52% | 50.60% | 0.0375 | 0.1172 | 0.5060 | 0.0356 | 1.3374 | 0.4482 | 10,966 | 14.0466 | 0.4569 |
| GCN + BGE-M3 (rank fusion) | 0.85% | 7.63% | 23.14% | 33.77% | 0.0335 | 0.0862 | 0.3377 | 0.0317 | 1.6650 | 0.4228 | 11,655 | 13.9222 | 0.4366 |
| BGE-M3 Only | 1.35% | 7.57% | 19.69% | 27.73% | 0.0366 | 0.0788 | 0.2773 | 0.0351 | 1.9413 | 0.5172 | 10,905 | 13.8459 | 0.3713 |
| GCN Only | 0.38% | 3.46% | 14.12% | 22.91% | 0.0180 | 0.0530 | 0.2291 | 0.0163 | 1.1879 | 0.2607 | 11,685 | 14.3293 | 0.5135 |
| GraphSAGE Only | 0.17% | 1.22% | 4.96% | 8.72% | 0.0076 | 0.0199 | 0.0872 | 0.0061 | 1.3134 | 0.2517 | 11,628 | 14.2087 | 0.5313 |

Several patterns are visible in Table 2. First, the corrected hybrid variants occupy the top positions on Hit@10 and remain strong on MRR, NDCG, Recall, and MAP under the revised evaluation protocol. Second, BGE-M3 Only is still the strongest model on Hit@1, indicating that pure semantic retrieval can place the correct article first more often even when it falls behind on deeper cutoffs. Third, Node2Vec-style retrieval remains competitive at broader cutoffs such as Hit@100 and also preserves favorable novelty and ILD behavior relative to several alternatives. The model ranking is therefore not identical across all recommendation-quality measures.

**Cutoff-10 and Link-Discrimination Metrics** Table 3 adds a smaller-cutoff view together with ROC-AUC and PR-AUC. This table helps separate recommendation behavior near the top of the list from broad discrimination across positive and negative edges.

**Table 3.** Cutoff-10 ranking metrics and link-discrimination metrics on the revised protocol

| Model | H@10 | NDCG@10 | Recall@10 | MAP@10 | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|---|
| BGE pooled + Node2Vec (corrected) | 8.67% | 0.0409 | 8.67% | 0.0273 | 0.9563 | 0.9559 |
| BGE-M3 + Node2Vec (corrected) | 8.67% | 0.0413 | 8.67% | 0.0278 | 0.9582 | 0.9579 |
| BGE pooled + Node2Vec (uncorrected) | 7.83% | 0.0356 | 7.83% | 0.0229 | 0.9710 | 0.9716 |
| BGE-M3 + Node2Vec (uncorrected) | 7.83% | 0.0356 | 7.83% | 0.0229 | 0.9710 | 0.9716 |
| Node2Vec (uncorrected) | 7.81% | 0.0355 | 7.81% | 0.0229 | 0.9717 | 0.9714 |
| Node2Vec (corrected) | 7.81% | 0.0355 | 7.81% | 0.0228 | 0.9717 | 0.9714 |
| BGE-M3 Only | 7.64% | 0.0406 | 7.64% | 0.0298 | 0.8851 | 0.8886 |
| GCN + BGE-M3 (rank fusion) | 7.63% | 0.0355 | 7.63% | 0.0234 | 0.9356 | 0.9363 |
| GCN Only | 3.46% | 0.0161 | 3.46% | 0.0107 | 0.8954 | 0.8946 |
| GraphSAGE Only | 1.22% | 0.0060 | 1.22% | 0.0041 | 0.8691 | 0.8561 |

The cutoff-10 view refines the interpretation of the benchmark. The corrected hybrid variants remain strongest on Hit@10, while the non-pooled corrected hybrid is highest on NDCG@10 among the compared models. At the same time, BGE-M3 Only is strongest on MAP@10, showing that a purely semantic ranking can still be precision-competitive very near the top of the list. On the broader discrimination side, uncorrected structural and lightly fused models occupy the highest ROC-AUC and PR-AUC values. This indicates that the model family that works best for top-ranked recommendation is not necessarily the same one that works best for global edge discrimination.

**Discussion** Corrected hybrids provide the most balanced top-list behavior: the pooled model ties for the best Hit@10, while the non-pooled model is slightly stronger on MRR, NDCG, Recall, and MAP. Pooling balances semantic (1024-D) and structural (128-D) vectors, but its effect is modest; it is useful as an alignment device rather than the sole explanation for the hybrid gains. Degree correction shows a clearer interaction effect. Corrected and uncorrected pure Node2Vec are nearly identical, whereas corrected hybrids improve top-list metrics such as Hit@10 and MRR. The heuristic therefore appears to stabilize semantic-structural fusion more than it changes structural retrieval in isolation.

Semantic-only models remain competitive near the top of the ranking, while structural models are stronger for broad link discrimination. Model selection therefore depends on whether the objective is a short recommendation list or global edge separation. The benchmark layer is also important: reconstruction makes semantic and structural evidence available for the same article set, and the undirected protocol gives the compared model families a consistent structural target.

The evaluation nevertheless remains a proxy. Missing hyperlinks are not necessarily irrelevant, reader behavior is unobserved, and edge direction is discarded. Moreover, pooling and degree correction are heuristic rather than learned fusion mechanisms, and recovered text may contain link-related cues or title mentions.

## 5 Conclusion

We introduced Custom-WikiCS, a reconstructed, text-complete Wiki-CS benchmark, and evaluated semantic, structural, hybrid, and GNN-based article retrieval on it. Beyond the model comparison, the work closes a practical dataset gap: raw text and graph structure are aligned for every active node, allowing semantic representations to be regenerated as encoders evolve. The revised protocol therefore provides a common basis for comparing content-driven and topology-driven recommendations on the same articles.

Corrected semantic-structural hybrids provide the best balance for short recommendation lists, while semantic and structural models retain advantages on Hit@1, MAP@10, ROC-AUC, or PR-AUC. These results support complementarity rather than universal dominance: semantic evidence is valuable near the top of a ranking, while structural evidence remains effective for broader connectivity. The reconstructed benchmark makes that comparison reproducible with modern text encoders on a shared node set. Accordingly, hybrid retrieval is most compelling when the application prioritizes short ranked lists rather than global graph prediction in isolation.

Future work should examine broader Wikipedia domains, directed protocols, harder negatives, lexical and graph-heuristic baselines, and learned yet interpretable fusion.

**Availability of Data and Code** The codebase and dataset reconstruction artifacts used in this study are available at
`https://github.com/regsah/graph-final-project`.

**Disclosure of Interests.** The authors have no competing interests to declare.

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